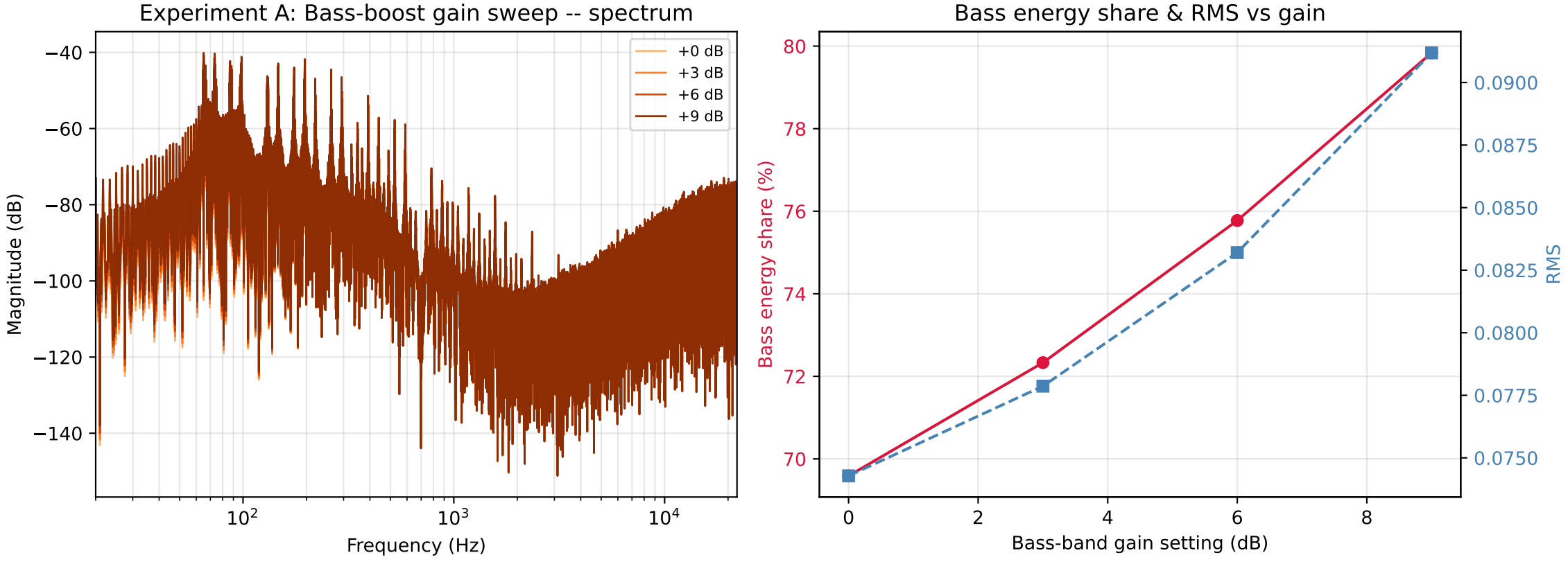
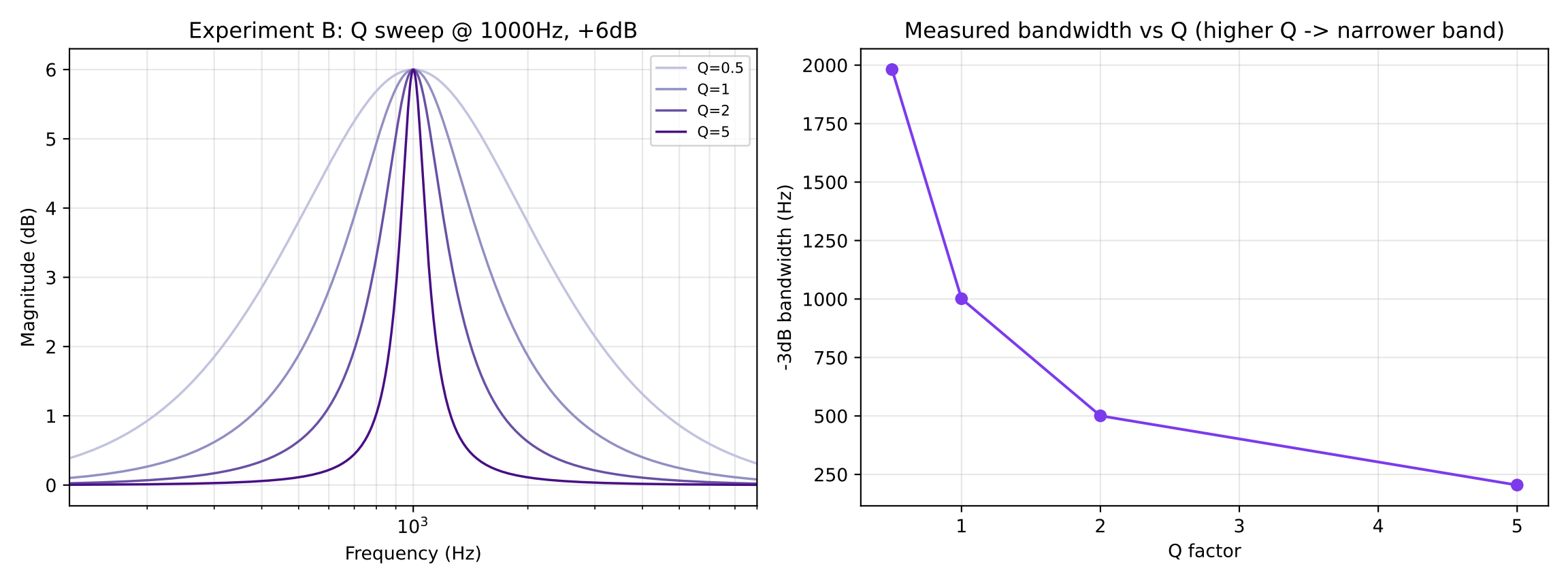
Experiment A: Bass Boost Gain Sweep

Gain (dB)	Bass energy %	RMS	Peak
+0	69.6%	0.0743	0.9000
+3	72.3%	0.0779	0.9019
+6	75.8%	0.0832	0.9091
+9	79.8%	0.0912	0.9193


Experiment A: Bass-boost gain sweep -- spectrum
Magnitude (dB)
−40
−60
−80
−100
−120
−140
10^2
10^3
10^4
Frequency (Hz)
+0 dB
+3 dB
+6 dB
+9 dB
Bass energy share & RMS vs gain
Bass energy share (%)
80
78
76
74
72
70
RMS
0.0900
0.0875
0.0850
0.0825
0.0800
0.0775
0.0750
0
2
4
6
8
Bass-band gain setting (dB)

Experiment B: Q Factor Sweep

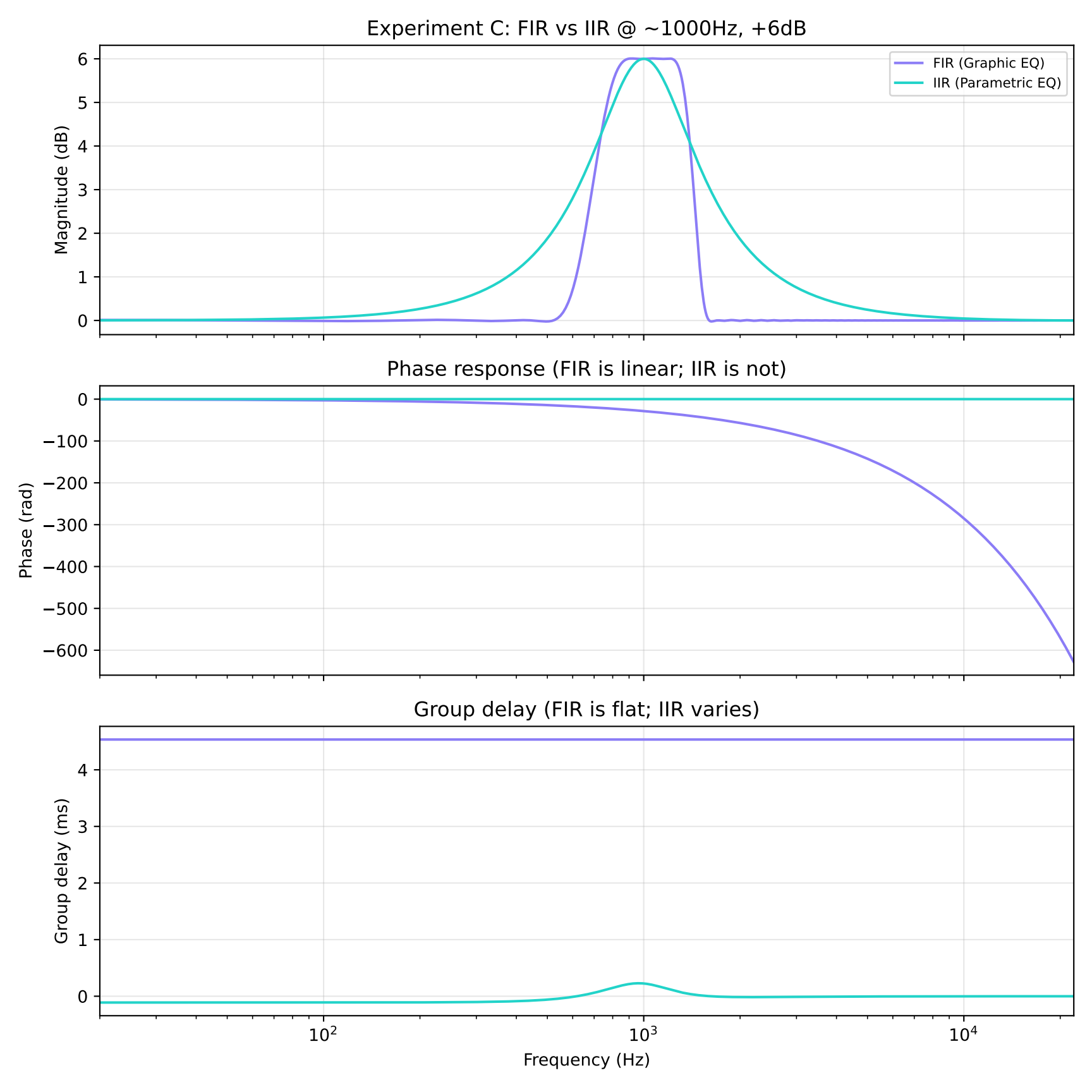
Q	Measured gain (dB)	-3dB Bandwidth (Hz)
0.5	+6.00	1981.1
1	+6.00	1001.3
2	+6.00	500.6
5	+6.00	204.6


Experiment B: Q sweep @ 1000Hz, +6dB
Q=0.5
Q=1
Q=2
Q=5
Magnitude (dB)
Frequency (Hz)
10^3
Measured bandwidth vs Q (higher Q -> narrower band)
-3dB bandwidth (Hz)
Q factor

Experiment C: FIR vs IIR

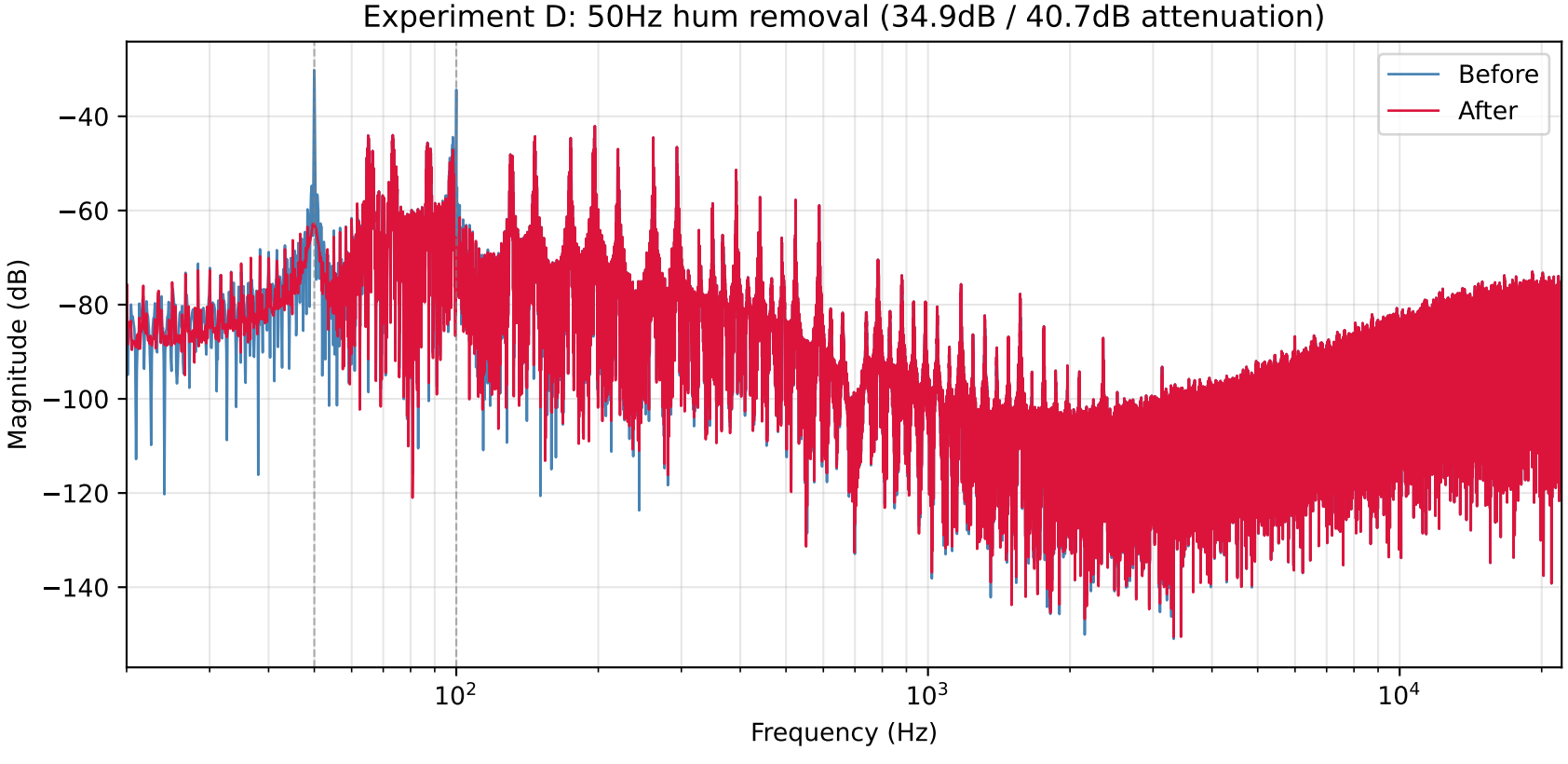
```
Target: 1000Hz, +6dB (graphic band used: 1000Hz)

FIR -- coefficients: 401, processing time: 338.82 ms
IIR -- coefficients: 6, processing time: 2.82 ms
```


Experiment C: FIR vs IIR @ ~1000Hz, +6dB
FIR (Graphic EQ)
IIR (Parametric EQ)
Magnitude (dB)
6
5
4
3
2
1
0
Phase response (FIR is linear; IIR is not)
Phase (rad)
0
−100
−200
−300
−400
−500
−600
Group delay (FIR is flat; IIR varies)
Group delay (ms)
4
3
2
1
0
10^2
10^3
10^4
Frequency (Hz)

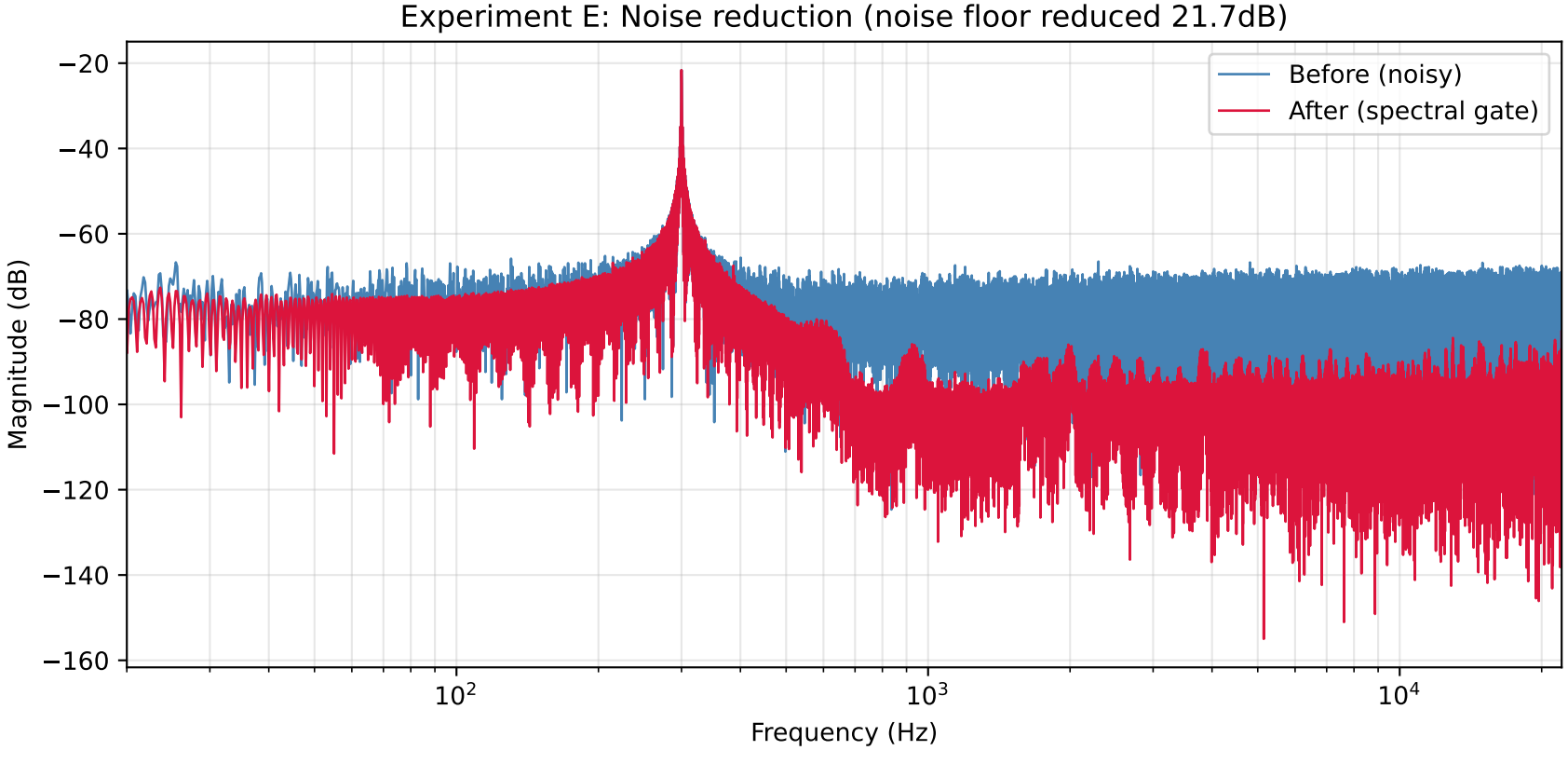
Experiment D: 50Hz Hum Removal

```
Fundamental (50Hz) attenuation : 34.9 dB
2nd harmonic (100Hz) attenuation: 40.7 dB
```


Experiment D: 50Hz hum removal (34.9dB / 40.7dB attenuation)
Before
After
Magnitude (dB)
−40
−60
−80
−100
−120
−140
10^2
10^3
10^4
Frequency (Hz)

Experiment E: Noise Reduction

```
Noise-only RMS before: 0.0499
Noise-only RMS after : 0.0041
Noise floor reduction: 21.7 dB
```


Experiment E: Noise reduction (noise floor reduced 21.7dB)
Before (noisy)
After (spectral gate)
Magnitude (dB)
−20
−40
−60
−80
−100
−120
−140
−160
10^2
10^3
10^4
Frequency (Hz)